

5. GIT BRANCHES: A BOOKMARK, NOT A COPY

Try an idea without touching what works

ANALOGY: THE FAMILY RECIPE BOOK



Mila

1 I want to try the soup with chili.

3 I'll use my own bookmark and new pages.

2 Not in the book we all cook from!

4 Fine. If it's good, we keep the pages.



Grandpa

Nobody copies the book. They agree on this:

- each new page says which page came before it
- a bookmark only marks the page you are on
- good soup: the new pages join the family book
- bad soup: throw the bookmark away

A branch is the bookmark: a name that points at one commit. Creating it copies no files.

THE SAME STORY IN GIT

	RECIPE BOOK	GIT
PAGE	one written page of the book	commit : a snapshot of all files + a link to its parent(s)
BOOKMARK	a small strip that marks a page	branch : a movable name pointing at one commit
FAMILY BOOKMARK	the one everyone cooks from	main , the usual name of the default branch
YOUR PLACE	the bookmark you are working from	HEAD : points at the branch you are on

LOOK FOR YOURSELF

```
$ cat .git/HEAD
ref: refs/heads/main
HEAD just names the branch you are on.

$ cat .git/refs/heads/main
d7b9e90f1dac36702a0ff2e6c705db70c0ce6259
The whole branch: one commit ID. (Git may also pack refs
into one file.)
```

ONE EXPERIMENT, START TO FINISH

1. NEW BOOKMARK

`git switch -c feature`
new branch here, HEAD moves to it



2. WRITE PAGES

`git add .`
`git commit -m "chili"`
the bookmark moves on



3. BACK TO FAMILY

`git switch main`
files return to main's snapshot



4. JOIN THE PAGES

`git merge feature`
main now has the chili

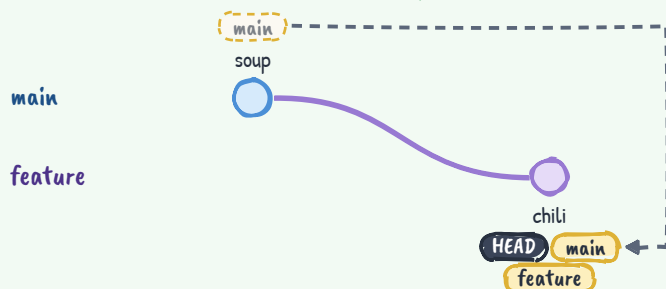


5. DROP THE BOOKMARK

`git branch -d feature`
refuses if unmerged; `-D` forces

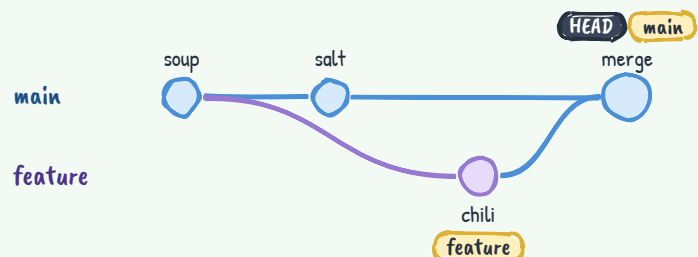


MERGE, CASE 1: FAST-FORWARD



Main had nothing new: **main** just slides to chili. No new commit.

MERGE, CASE 2: A MERGE COMMIT



Both moved on: a new commit with **two parents**. See it: `git log --graph`



CONFLICT: Both sides changed the same lines. Git marks them with `<<<<<<` and waits for a human to choose.



REMEMBER:

A branch is a bookmark on a commit, not a copy of your files. Make them freely, merge the good ones.